

Finite Difference Method – Rectangular Plate

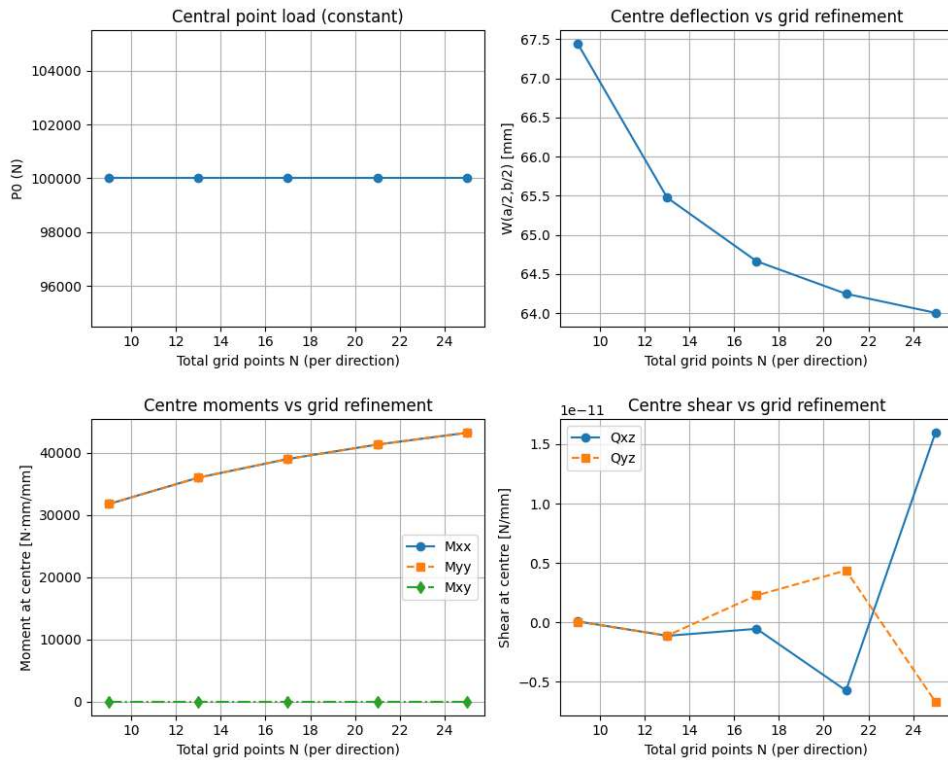
BC: 3 sides simply supported ($w=0$), bottom edge treated as clamped in interpretation

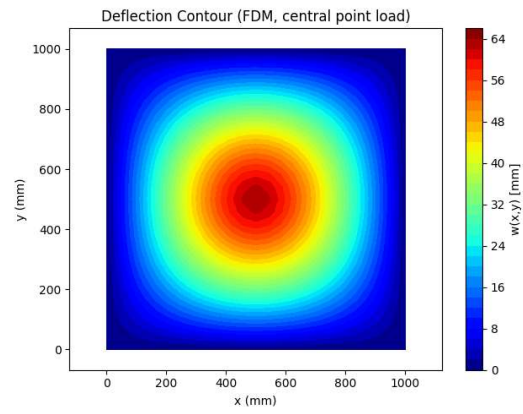
Load: central point load P_0 (N)

Convergence at plate centre ($x=a/2, y=b/2$)

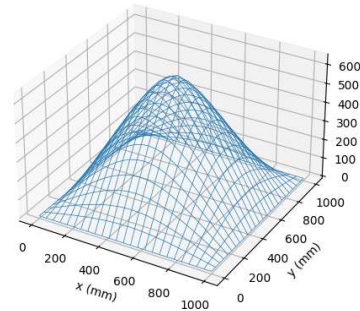
N = total grid points in each direction (including boundaries)

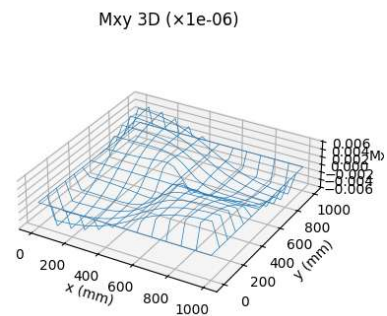
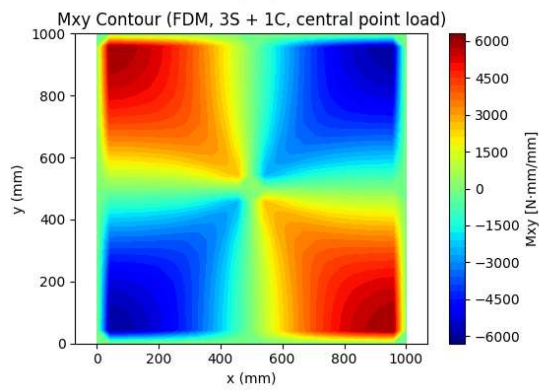
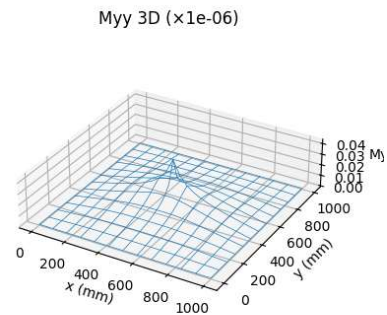
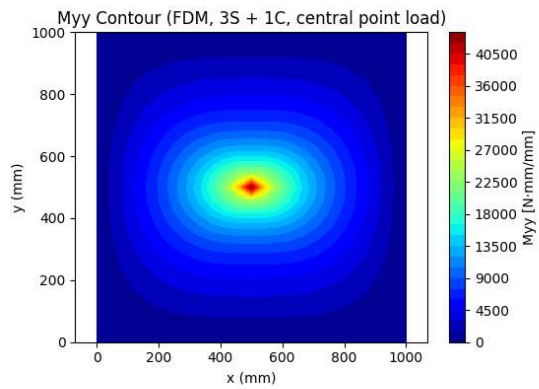
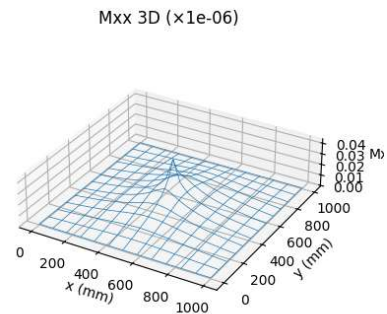
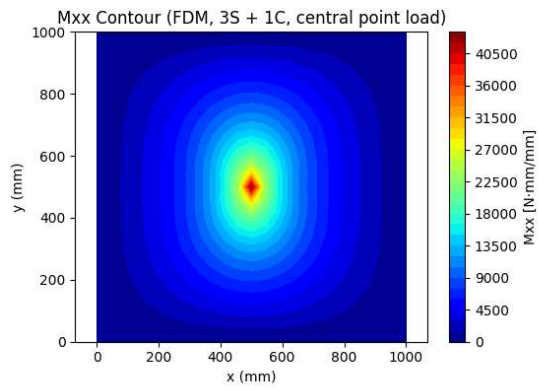
N	P_0	W_{xy}	M_{xx}	M_{yy}	M_{xy}	Q_{xz}	Q_{yz}
9	100000.00 N	6.7439e+01	3.1783e+04	3.1783e+04	-8.7451e-12	7.2760e-14	5.0978e-14
13	100000.00 N	6.5474e+01	3.6017e+04	3.6017e+04	-0.0000e+00	-1.1375e-12	-1.1243e-12
17	100000.00 N	6.4664e+01	3.9006e+04	3.9006e+04	5.8301e-11	-5.5297e-13	2.2652e-12
21	100000.00 N	6.4249e+01	4.1321e+04	4.1321e+04	-9.1095e-12	-5.7498e-12	4.3664e-12
25	100000.00 N	6.4006e+01	4.3210e+04	4.3210e+04	1.3118e-11	1.5958e-11	-6.6953e-12

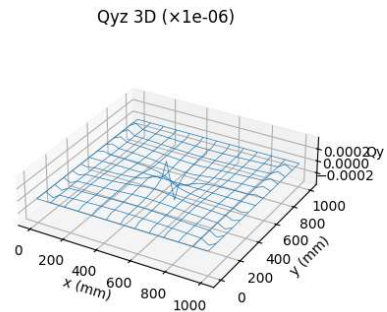
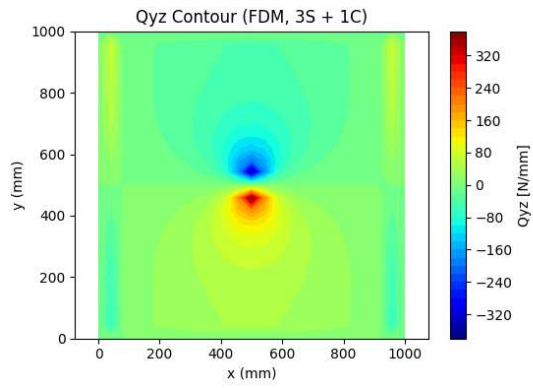
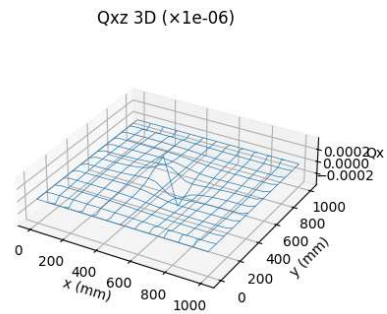
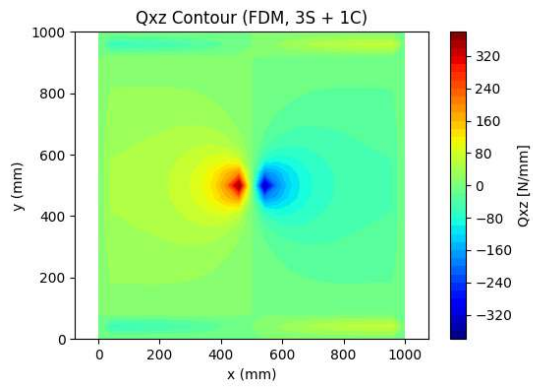




3D Deflection ($\times 10.0$) - 3S + 1C ($y=0$ clamped)







From <<http://localhost:56928/doc>>